

Research Seminars on Diffusion Models

RECORDING ARCHIVE · 2023–2026

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[Seminar Telegram group ↗](#)

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151 seminars · Archive as of September 2026

2026

16.09.26 Watch recording Access code cK*UUF1i	Diffusion-based speculative decoding, continued. — DART — DDTree — TiDAR — Orthrus Speakers: Arseniy Rybakov
09.09.26 Watch recording Access code x.Ewjv&6	A survey of diffusion drafters for LLMs — DFlash — DFlare — DSpark : adding a small RNN on top of the drafter — DFlash2 : PathSelector and other ways to address inconsistencies in parallel token decoding — DART : PathSelection using an n-gram model — DDTree : how TreeAttention in the verifier can help when the inference engine operates at low concurrency — ReTrace : reusing tokens rejected by the verifier Finally, approaches in which the drafter and target model share the same backbone: TiDAR and Orthrus Speakers: Arseniy Rybakov Additional source links: [1] · [2] · [3] · [4] · [5] · [6] · [7]
02.09.26 Watch recording Access code &4FBi@7e	Recirculation Speakers: Nikita Starodubcev

26.08.26 Watch recording Access code KYL@01Ga	Continuous diffusion language models Speakers: Denis Kuznedelev
19.08.26 Watch recording Access code g*6v4RVU	Neural Thickets: Diverse Task Experts Are Dense Around Pretrained Weights WaiT for the Signal: Simple Frequency-Aware Flow-Matching Speakers: Nikita Starodubcev
12.08.26 Watch recording Access code hUp7\$Y6K	Verbalizable Representations Form a Global Workspace in Language Models Speakers: Nikita Starodubcev
05.08.26 Watch recording Access code G=3nX.ev	Explorative Modeling: Unlocking a Third Pretraining Axis and End-to-End Generation Speakers: @dbaranchuk
29.07.26 Watch recording Access code sz4Q83&%	Stop Training for the Worst: Progressive Unmasking Accelerates Masked Diffusion Training Speakers: Artem Makoyan
22.07.26 Watch recording Access code BKf\+6Ww9	The Flexibility Trap: Rethinking the Value of Arbitrary Order in Diffusion Language Models DMax: Aggressive Parallel Decoding for dLLMs Speakers: Denis Kuznedelev @dbaranchuk
15.07.26 Watch recording Access code ^5tyxwM6	Self-conditioned Flow Map Language Models via Fixed-point Flows Set Diffusion: Interpolating Token Orderings Between Autoregression and Diffusion for Fast and Flexible Decoding
08.07.26 Watch recording Access code H%2Azk\$T	The Truth Lies Somewhere in the Middle (of the Generated Tokens) DiLaDiff: Distilled Latent-Augmented Diffusion for Language Modeling Where Should Diffusion Enter a Language Model? Geometry-Guided Hidden-State Replacement

01.07.26 Watch recording Access code E3W@04Dp	Posterior Refinement: Fast Language Generation via Any-Order Flow Maps
24.06.26 Watch recording Access code 4QWx^?S3	LangFlow Language Modeling with Hyperspherical Flows Diffusion forcing Discrete Diffusion Forcing (D2F): dLLMs Can Do Faster-Than-AR Inference
17.06.26 Watch recording Access code 6h6RpC@q	Flow Map Language Models: One-step Language Modeling via Continuous Denoising LangFlow: Continuous Diffusion Rivals Discrete in Language Modeling Speakers: @dbaranchuk @iasudakov
10.06.26 Watch recording Access code 2n3.rsib	Sparse Forcing: Native Trainable Sparse Attention for Real-time Autoregressive Diffusion Video Generation FLARE: Diffusion for Hybrid Language Model Speakers: @fvelikon Denis Kuznedelev
03.06.26 Watch recording Access code 26*dy.Bg	FreeU Stable Flow FreeFlux TACA Calibri Speakers: Danil Tokhchukov
22.05.26 Watch recording Access code Cav%*=?1	Improved Baselines with Representation Autoencoders
20.05.26 Watch recording Access code 9wjrr9!&	Byte Latent Transformer Fast Byte Latent Transformer Speakers: Denis Kuznedelev

13.05.26 Watch recording Access code +uv+^5Wi	ELF: Embedded Language Flows Representation Fréchet Loss for Visual Generation Speakers: @dbaranchuk Nikita Starodubcev
15.04.26 Watch recording Access code #g#LK+i6	ELT: Elastic Looped Transformers for Visual Generation Cubic Discrete Diffusion: Discrete Visual Generation on High-Dimensional Representation Tokens
01.04.26 Watch recording Access code xJHgFh!8	Foveated Diffusion:Efficient Spatially Adaptive Image and Video Generation Speakers: @kstruminsky
25.03.26 Watch recording Access code zmy5Y\$y9	Beyond Single Tokens: Distilling Discrete Diffusion Models via Discrete MMD Speakers: Denis Kuznedelev
20.03.26 Watch recording Access code c8K@TH6e	Representation Alignment for Just Image Transformers is not Easier than You Think PixelDiT: Pixel Diffusion Transformers for Image Generation PixelGen: Pixel Diffusion Beats Latent Diffusion with Perceptual Loss Speakers: Nikita Starodubcev @free001style
18.03.26 Watch recording Access code cQ28kz!L	Reviving ConvNeXt for Efficient Convolutional Diffusion Models One Model, Many Budgets: Elastic Latent Interfaces for Diffusion Transformers Speakers: @free001style @s-kastryulin
06.03.26 Watch recording Access code a*p^N6c9	Self-Supervised Flow Matching for Scalable Multi-Modal Synthesis Laminating Representation Autoencoders for Efficient Diffusion (related to https://arxiv.org/pdf/2512.07829) Improving Reconstruction of Representation Autoencoder Speakers: @free001style Additional source links: [1]

04.03.26 Watch recording Access code Bc0.L9mh	Sparse Video-Gen: Accelerating Video Diffusion Transformers with Spatial-Temporal Sparsity Sparse VideoGen2: Accelerate Video Generation with Sparse Attention via Semantic-Aware Permutation Speakers: Denis Kuznedelev
27.02.26 Watch recording Access code t%p=f86C	Register and [CLS]tokens induce a decoupling of local and global features in large ViTs SINDER: Repairing the Singular Defects of DINOv2 Revisiting [CLS] and Patch Token Interaction in Vision Transformers Speakers: Nikita Starodubcev
25.02.26 Watch recording Access code B89f^mLi	Unified Latents (UL): How to train your latents Speakers: Denis Kuznedelev
20.02.26 Watch recording Access code @K@!UbU0	Transition Matching: Scalable and Flexible Generative Modeling Exploring the Design Space of Transition Matching PixelDiT: Pixel Diffusion Transformers for Image Generation PixelGen: Pixel Diffusion Beats Latent Diffusion with Perceptual Loss Speakers: @iasudakov @free001style
18.02.26 Watch recording Access code ^Gyt#Y5R	FLUX.2: Analyzing and Enhancing the Latent Space of FLUX – Representation Comparison
11.02.26 Watch recording Access code sS0!gv7J	Generative Modeling via Drifting Speakers: @iasudakov @free001style
04.02.26 Watch recording Access code .B!VTz6=	MeanCache: From Instantaneous to Average Velocity for Accelerating Flow Matching Inference One-step Latent-free Image Generation with Pixel Mean Flows Speakers: Nikita Starodubcev

28.02.26 Watch recording Access code YUNsm.3&	Scaling Text-to-Image Diffusion Transformers with Representation Autoencoders
16.01.26 Watch recording Access code da?*e98j	There is No VAE: End-To-End Pixel-Space Generative Modeling via Self-Supervised Pre- training One Layer Is Enough: Adapting Pretrained Visual Encoders for Image Generation Toward Diffusible High-Dimensional Latent Spaces: A Frequency Perspective
14.01.26 Watch recording Access code v3dg8\+T0	1. NextFlow: Unified Sequential Modeling Activates Multimodal Understanding and Generation 2. Unified Text-Image Generation with Weakness-Targeted Post-Training Speakers: Denis Kuznedelev @troeshestoval

2025

24.12.25

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Access code
L\$ixmw7

Group Diffusion: Enhancing Image Generation by Unlocking Cross-Sample Collaboration

FreeFlow: Flow Map Distillation Without Data

17.12.25

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%kFIE4u7

Why Diffusion Models Don't Memorize: The Role of Implicit Dynamical Regularization in Training (Denis Kuznedelev)

What matters for Representation Alignment: Global Information or Spatial Structure? (@dbranchuk)

12.12.25

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Access code
0QaSl#z*

Align Your Flow: Scaling Continuous-Time Flow Map Distillation (Nikita Starodubcev)

10.12.25

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Access code
bC9k8Hw?

Improved Mean Flows: On the Challenges of Fastforward Generative Models

03.12.25

[Watch recording](#)

Access code
v+0#F=M8

AlphaFlow: Understanding and Improving MeanFlow Models

CMT: Mid-Training for Efficient Learning of Consistency, Mean Flow, and Flow Map Models

Understanding, Accelerating, and Improving MeanFlow Training

26.11.25

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Access code
Q\$l%!T5^

Latent Diffusion Model without Variational Autoencoder (Nikita Starodubcev)

05.11.25 Watch recording Access code W.J3vLq6	Diffusion-LM Improves Controllable Text Generation Self-conditioned Embedding Diffusion for Text Generation Continuous Diffusion for Categorical Data Latent Diffusion for Language Generation PLANNER: Generating Diversified Paragraph via Latent Language Diffusion Model Empowering Diffusion Models on the Embedding Space for Text Generation TEncDM: Understanding the Properties of the Diffusion Model in the Space of Language Models Compressed and Smooth Latent Space for Text Diffusion Modeling Speaker: @meshchaninov
01.11.25 Watch recording Access code 8TIp+185	Selective Underfitting in Diffusion Models Locality in Image Diffusion Models Emerges from Data Statistics Understanding Generalization in Diffusion Models via Probability Flow Distance
31.10.25 Watch recording Access code ++!SE7k*	Understanding Generalizability of Diffusion Models Requires Rethinking the Hidden Gaussian Structure Towards a Mechanistic Explanation of Diffusion Model Generalization Selective Underfitting in Diffusion Models Speaker: Nikita Starodubcev
29.10.25 Watch recording Access code \$n#a6g2m	Planned Diffusion (Denis Kuznedelev) Continuously Augmented Discrete Diffusion model for Categorical Generative Modeling (@free001style) Coevolutionary Continuous Discrete Diffusion: Make Your Diffusion Language Model a Latent Reasoner (@free001style)
22.10.25 Watch recording Access code A=j@2i&2	Diffusion Transformers with Representation Autoencoders (@mikhailkolt)
18.10.25 Watch recording Access code yFEv1E\$+	Nikita Starodubcev continued the discussion of the physical foundations and the connection between PFGM and diffusion, from a perspective different from PFGM++. Additional source links: [1]

17.10.25 Watch recording Access code F0*QrrPj	@d-shlenskii will finish presenting his distillation paper, and Nikita Starodubcev will discuss the physics behind the fundamental equations of PFGM. Additional source links: [1] · [2]
15.10.25 Watch recording Access code #%Ayb962	Poisson Flow Generative Models PFGM++: Unlocking the Potential of Physics-Inspired Generative Models Overclocking Electrostatic Generative Models Field Matching: an Electrostatic Paradigm to Generate and Transfer Data Interaction Field Matching: Overcoming Limitations of Electrostatic Models Speaker: @d-shlenskii
08.10.25 Watch recording Access code 07sBr+Gj	Poisson Flow Generative Models (@daniil_shlenskii) PFGM++: Unlocking the Potential of Physics-Inspired Generative Models (@daniil_shlenskii) Overclocking Electrostatic Generative Models (@daniil_shlenskii) Field Matching: an Electrostatic Paradigm to Generate and Transfer Data (@daniil_shlenskii) Interaction Field Matching: Overcoming Limitations of Electrostatic Models (@daniil_shlenskii) To be continued the following week.
01.10.25 Watch recording Access code #V=2fsC0	Transition Models: Rethinking the Generative Learning Objective (Nikita Starodubcev) Understand Before You Generate: Self-Guided Training for Autoregressive Image Generation (@iasudakov)
26.09.25 Watch recording Access code xtRg4u^k	GScreeam: Learning 3D Geometry and Feature Consistent Gaussian Splatting for Object Removal Speakers: @mvkolos and Arseniy Rybakov Additional source links: [1] · [2]
12.09.25 Watch recording Access code dwa3D\$5#	RawNeRF (@kstruminsky)

05.09.25 Watch recording Access code #JS9p0r!	PF3plat: Pose-Free Feed-Forward 3D Gaussian Splatting No Pose at All: Self-Supervised Pose-Free 3D Gaussian Splatting from Sparse Views Speaker: @mishanaliev
29.08.25 Watch recording Access code KC0hM%=s	Splatt3R: Zero-shot Gaussian Splatting from Uncalibrated Image Pairs Uni3R: Unified 3D Reconstruction and Semantic Understanding via Generalizable Gaussian Splatting from Unposed Multi-View Images Speaker: @mishanaliev
27.08.25 Watch recording Access code \$ivk8W2e	Diffusion Beats Autoregressive in Data-Constrained Settings (Denis Kuznedelev) Diffusion Language Models are Super Data Learners (Denis Kuznedelev)
22.08.25	Next Best View in 3D computer vision (@mishanaliev) Additional source links: [1]
08.08.25 Watch recording Access code 0u#iP#fH	UniRelight (@kstruminsky)
06.08.25 Watch recording Access code f1JWX@?R	Radial Attention: $O(n \log n)$ Sparse Attention with Energy Decay for Long Video Generation Sparse VideoGen: Accelerating Video Diffusion Transformers with Spatial-Temporal Sparsity
01.08.25 Watch recording Access code w&90DgN6	Your Diffusion Model is Secretly a Zero-Shot Classifier Generative Classifiers Avoid Shortcut Solutions Chroma-VAE: Mitigating Shortcut Learning with Generative Classifiers Intriguing properties of generative classifiers Generative or Discriminative? Revisiting Text Classification in the Era of Transformers
23.07.25 Watch recording Access code b*q0Vcsu	Vision as a Dialect: Unifying Visual Understanding and Generation via Text-Aligned Representations (Denis Kuznedelev) Image Generation from Contextually-Contradictory Prompts (@dbaranchuk)

16.07.25 Watch recording Access code zC&jbWh1	Why Is Spatial Reasoning Hard for VLMs? An Attention Mechanism Perspective on Focus Areas (@s-kastryulin) Vision-Language-Vision Auto-Encoder: Scalable Knowledge Distillation from Diffusion Models (@s-kastryulin)
11.07.25 Watch recording Access code p@dr15T	Text Embedding is Not All You Need: Attention Control for Text-to-Image Semantic Alignment with Text Self-Attention Maps Linguistic Binding in Diffusion Models: Enhancing Attribute Correspondence through Attention Map Alignment Towards Understanding the Working Mechanism of Text-to-Image Diffusion Model Normalized Attention Guidance: Universal Negative Guidance for Diffusion Models Attend-and-Excite: Attention-Based Semantic Guidance for Text-to-Image Diffusion Models Why Is Spatial Reasoning Hard for VLMs? An Attention Mechanism Perspective on Focus Areas Speaker: Nikita Starodubcev
09.07.25 Watch recording Access code R18Ss0?F	Di[M]O: Distilling Masked Diffusion Models into One-step Generator (@nikitagushchin) Beyond Autoregression: Fast LLMs via Self-Distillation Through Time (@nikitagushchin) The Diffusion Duality (@nikitagushchin)
02.07.25 Watch recording Access code b??qZ86B	Di[M]O: Distilling Masked Diffusion Models into One-step Generator Beyond Autoregression: Fast LLMs via Self-Distillation Through Time The Diffusion Duality Speaker: @nik-gushchin
27.06.25 Watch recording Access code 7MPj&01d	Diffusion as Shader: 3D-aware Video Diffusion for Versatile Video Generation Control Voyager: Long-Range and World-Consistent Video Diffusion for Explorable 3D Scene Generation Speaker: @mishanaliev
27.06.25 Watch recording Access code #lXu&7R4	Token-Shuffle: Towards High-Resolution Image Generation with Autoregressive Models RandAR: Decoder-only Autoregressive Visual Generation in Random Orders

18.06.25 Watch recording Access code A98r7!cw	Self Forcing: Bridging the Train-Test Gap in Autoregressive Video Diffusion Autoregressive Adversarial Post-Training for Real-Time Interactive Video Generation Training-free Guidance in Text-to-Video Generation via Multimodal Planning and Structured Noise Initialization VLIPP: Towards Physically Plausible Video Generation with Vision and Language Informed Physical Prior Speaker: @free001style
11.06.25 Watch recording Access code m7#^54aW	Are Unified Vision-Language Models Necessary: Generalization Across Understanding and Generation Concept Reachability in Diffusion Models: Beyond Dataset Constraints A Comprehensive Study of Decoder-Only LLMs for Text-to-Image Generation Speaker: @s-kastryulin
06.06.25 Watch recording Access code 1T!l4UP\$	1. Physically based rendering: the rendering equation and its key components, including radiance fields, scattering functions, and the underlying integrals. 2. Simplifying the rendering equation to obtain popular relighting techniques for real-time graphics. Approximations using spherical harmonics (https://github.com/joaxkal/lumigauss/tree/main) and mean-field-style methods for PBR materials (based on https://asparagus15.github.io/GaussianShader.github.io/). 3. Time permitting, an introduction to inverse rendering. The first two topics cover rendering an image from a 3D model; how can we solve the inverse problem? Proposed reading: https://cseweb.ucsd.edu/~ravir/6998/papers/invrend.pdf and https://benattal.github.io/flash-cache/. Speaker: @kstruminsky Additional source links: [1] · [2] · [3]
04.06.25 Watch recording Access code 04+3ziTz	1. REPA-E: Unlocking VAE for End-to-End Tuning with Latent Diffusion Transformers (@iasudakov) 2. VideoREPA: Learning Physics for Video Generation through Relational Alignment with Foundation Models (@free001style) 3. No Other Representation Component Is Needed: Diffusion Transformers Can Provide Representation Guidance by Themselves (@free001style)
28.05.25 Watch recording Access code 2w&YFBus	Mean Flows for One-step Generative Modeling (@dbaranchuk)

07.05.25 Watch recording Access code 5EJz0z\$T	1. From Slow Bidirectional to Fast Autoregressive Video Diffusion Models (@free001style) 2. Video Diffusion Models are Training-free Motion Interpreter and Controller (@iasudakov)
30.04.25 Watch recording Access code *8#vctmp	Eliminating Oversaturation and Artifacts of High Guidance Scales in Diffusion Models (@dbaranchuk) Autoregressive Distillation of Diffusion Transformers (Denis Kuznedelev)
23.04.25 Watch recording Access code 3Zzc&8x!	I Think, Therefore I Diffuse: Enabling Multimodal In-Context Reasoning in Diffusion Models (Nikita Starodubcev) PixelFlow: Pixel-Space Generative Models with Flow (Nikita Starodubcev)
18.04.25 Watch recording Access code c5*Ik#f1	Using semantic features from pretrained models to accelerate the training of diffusion models and VAEs. Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think (@dbaranchuk) U-REPA: Aligning Diffusion U-Nets to ViTs (@dbaranchuk) Reconstruction vs. Generation: Taming Optimization Dilemma in Latent Diffusion Models (Nikita Starodubcev) DDT: Decoupled Diffusion Transformer (Nikita Starodubcev)
09.04.25 Watch recording Access code fjPths7.	An introduction to recent developments in multimodal models. Unified Autoregressive Visual Generation and Understanding with Continuous Tokens (@dbaranchuk) Transfusion: Predict the Next Token and Diffuse Images with One Multi-Modal Model (@s-kastryulin)
2.04.25 Watch recording Access code iRT76t=#	Fractal Generative Models (@dbaranchuk)
25.03.25 Watch recording Access code K6*.Z4Ud	The Best of Both Worlds: Integrating Language Models and Diffusion Models for Video Generation (@s-kastryulin) SANA 1.5: Efficient Scaling of Training-Time and Inference-Time Compute in Linear Diffusion Transformer (@s-kastryulin)

21.03.25 Watch recording Access code \$6J^mMu*	Edify Image: High-Quality Image Generation with Pixel Space Laplacian Diffusion Models (Nikita Starodubcev) SANA-Sprint: One-Step Diffusion with Continuous-Time Consistency Distillation (Nikita Starodubcev)
19.03.25 Watch recording Access code UtV7.0rW	Distilling Diversity and Control in Diffusion Models (@dbaranchuk) Is Noise Conditioning Necessary for Denoising Generative Models? (@dbaranchuk) Do generative video models understand physical principles? (@dbaranchuk)
12.03.25 Watch recording Access code t.Jv=u3d	FlexiDiT: Your Diffusion Transformer Can Easily Generate High-Quality Samples with Less Compute (@dbaranchuk) Beyond Next-Token: Next-X Prediction for Autoregressive Visual Generation (@dbaranchuk)
26.02.25 Watch recording Access code SV15%q9G	Improving the Diffusability of Autoencoders (@s-kastryulin) EQ-VAE: Equivariance Regularized Latent Space for Improved Generative Image Modeling (@s- kastryulin)
21.02.25 Watch recording Access code w?\$3qXRS	(Mishan Aliev) How Gaussian splatting represents 3D models.
19.02.25 Watch recording Access code V9F%@B.%	Step-Video-T2V Technical Report: The Practice, Challenges, and Future of Video Foundation Model (@s-kastryulin)
12.02.25 Watch recording Access code yR2Jc2*+	SliderSpace: Decomposing the Visual Capabilities of Diffusion Models (@dbaranchuk)
7.02.25 Watch recording Access code wv=G+4rv	A3D: Does Diffusion Dream about 3D Alignment? (@daniilssel)

5.02.25 Watch recording Access code 3+YNVpc@	Compositional Abilities Emerge Multiplicatively: Exploring Diffusion Models on a Synthetic Task (Denis Kuznedelev) Visual Autoregressive Modeling for Image Super-Resolution (Denis Kuznedelev)
24.01.25 Watch recording Access code h=ksX1jH	ϵ -VAE: Denoising as Visual Decoding (Nikita Starodubcev) TokenVerse: Versatile Multi-concept Personalization in Token Modulation Space (Nikita Starodubcev) Large-Scale Text-to-Image Model with Inpainting is a Zero-Shot Subject-Driven Image Generator (Nikita Starodubcev)
22.01.25 Watch recording Access code qyw=30^V	Inference-Time Scaling for Diffusion Models beyond Scaling Denoising Steps (Denis Kuznedelev)
10.01.25 Watch recording Access code \$B4=c fuW	Efficient Scaling of Diffusion Transformers for Text-to-Image Generation (@dbaranchuk) Flowing from Words to Pixels: A Framework for Cross-Modality Evolution (@dbaranchuk) FlowAR: Scale-wise Autoregressive Image Generation Meets Flow Matching (@dbaranchuk)

2024

28.12.24

[Watch recording](#)

Access code
FH?2x&s@

[Diffusion Meets Flow Matching: Two Sides of the Same Coin \(Denis Kuznedelev \)](#)

24.12.24

[Watch recording](#)

Access code
YXV3qT2%

[Meta 3D TextureGen: Fast and Consistent Texture Generation for 3D Objects \(Mishan Aliev\)](#)

[MatAtlas: Text-driven Consistent Geometry Texturing \(Mishan Aliev\)](#)

[MVPaint: Synchronized Multi-View Diffusion for Painting Anything 3D \(Mishan Aliev\)](#)

18.12.24

[Watch recording](#)

Access code
PQ\$1Jy@p

[Randomized Autoregressive Visual Generation \(@notantonvoron \)](#)

[Collaborative Decoding Makes Visual Auto-Regressive Modeling Efficient \(@notantonvoron \)](#)

13.12.24

[Watch recording](#)

Access code
tU1^?5r&

[One-Step Diffusion Distillation through Score Implicit Matching \(@daniilssel \)](#)

11.12.24

[Watch recording](#)

Access code
4u%+y3bQ

[Infinity: Scaling Bitwise AutoRegressive Modeling for High-Resolution Image Synthesis \(Denis Kuznedelev \)](#)

04.12.24

[Watch recording](#)

Access code
8#V^ps!3

[On Improved Conditioning Mechanisms and Pre-training Strategies for Diffusion Models \(@s-kastryulin \)](#)

27.11.24

[Watch recording](#)

Access code
Xt\$us=K1

[A survey of recent text-to-video work and new VAEs for text-to-image+video \(@s-kastryulin \)](#)

[JVID: Joint Video-Image Diffusion for Visual-Quality and Temporal-Consistency in Video Generation](#)
[Pyramidal Flow Matching for Efficient Video Generative Modeling](#)
[Cosmos Tokenizer: A suite of image and video neural tokenizers](#)
[REDUCIO! Generating 1024×1024 Video within 16 Seconds using Extremely Compressed Motion Latents](#)
[Movie Gen: A Cast of Media Foundation Models](#)

20.11.24 Watch recording Access code FJ8.rH+V	Adversarial Schrodinger Bridge Matching (@daniilseI)
6.11.24 Watch recording Access code 440nJFy!	Multi-student Diffusion Distillation for Better One-step Generators (Nikita Starodubcev) Unpacking SDXL Turbo: Interpreting Text-to-Image Models with Sparse Autoencoders (Nikita Starodubcev)
30.10.24 Watch recording Access code j1M.1s&c	Simplifying, Stabilizing and Scaling Continuous-Time Consistency Models (Denis Kuznedelev)
21.10.24 Watch recording Access code \$+UW7=u@	Deep Compression Autoencoder for Efficient High-Resolution Diffusion Models (@s-kastryulin) Sample what you cant compress (@s-kastryulin)
16.10.24 Watch recording	Mixture of Efficient Diffusion Experts Through Automatic Interval and Sub-Network Selection
16.10.24 Watch recording	M-VAR: Decoupled Scale-wise Autoregressive Modeling for High-Quality Image Generation
11.10.24 Watch recording Access code Pq7JF^^M	Movie Gen: A Cast of Media Foundation Models (Denis Kuznedelev) Part 2
04.10.24 Watch recording Access code 6Eqpf0=y	Movie Gen: A Cast of Media Foundation Models (Denis Kuznedelev) Part 1 Discussion of the architecture (autoencoder, text encoders, and diffusion model), training details, and data.

25.09.24 Watch recording Access code rSy@y9#q	Updates on prompt optimization for text-to-image diffusion, followed by methods for removing unwanted concepts from pretrained diffusion models. On Discrete Prompt Optimization for Diffusion Models (@s-kastryulin) Erasing Concepts from Diffusion Models (@s-kastryulin) R.A.C.E.: Robust Adversarial Concept Erasure for Secure Text-to-Image Diffusion Model (@s-kastryulin)
11.09.24 Watch recording Access code bRvM?Q7d	Step-aware Preference Optimization: Aligning Preference with Denoising Performance at Each Step (Nikita Starodubcev)
06.09.24 Watch recording Access code 2jzN%w?Z	(Sasha Korotin) An introduction to the emerging research area of training neural generative models for data-to-data tasks, such as image-to-image translation, using optimal transport (OT) and diffusion bridges. The first part motivates OT, particularly for unpaired domain transfer, and covers adversarial methods for learning OT. The second part introduces a special case of OT: the Schrödinger bridge (SB). It reviews SB properties, motivates its use in generative modeling, and surveys learning algorithms based on adversarial modeling and bridge matching, which are conceptually close to diffusion models.
28.08.24 Watch recording Access code ?#m*2tJa	Distilling Diffusion Models into Conditional GANs (@dbaranchuk)
21.08.24 Watch recording Access code ?dGG5+@D	TurboEdit: Instant text-based image editing (Nikita Starodubcev)
16.08.24 Watch recording Access code pN?u#^P2	TurboEdit: Text-Based Image Editing Using Few-Step Diffusion Models (Nikita Starodubcev) CFG++: Manifold-constrained Classifier Free Guidance for Diffusion Models (@dbaranchuk)
02.08.24 Watch recording Access code +UJ7ormS	(Nikita Morozov) GFlowNets: another family of generative models closely related to diffusion and reinforcement learning fine-tuning. Flow Network based Generative Models for Non-Iterative Diverse Candidate Generation Generative flow networks as entropy-regularized RL

31.07.24	Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution (Denis Kuznedelev)
26.07.24 Watch recording Access code ?u4YgqWJ	Text-to-3D generation and a recent advance in the field. CAT3D: Create Anything in 3D with Multi-View Diffusion Models (@kstruminsky)
24.07.24 Watch recording Access code jJF0G^D@	Scaling Diffusion Transformers to 16 Billion Parameters (Denis Kuznedelev) Rethinking FID: Towards a Better Evaluation Metric for Image Generation (Dmitry Baranchuk)
19.07.24 Watch recording Access code Um*sx5!r	(Grisha Bartosh) Grisha presents two of his papers on extending diffusion models to more expressive forward processes. Neural Diffusion Models Neural Flow Diffusion Models: Learnable Forward Process for Improved Diffusion Modelling
17.07.24 Watch recording Access code i7mWCdY!	(Vanya Shvechikov) Semi-implicit Variational Inference via Score Matching, plus a derivation of SiD Additional source links: [1]
12.07.24 Watch recording Access code Yq2U&gg+	Multistep Distillation of Diffusion Models via Moment Matching (Nikita Starodubcev)
10.07.24 Watch recording Access code @5b Passcode: &K6N^[@5b] (https://sta ff.yandex-team.ru/5b)	A continued discussion of SiD and related work to develop a more complete picture. EM Distillation for One-step Diffusion Models (Nikita Starodubcev) Additional source links: [1] · [2]
05.07.24 Watch recording Access code &zJyqp5P	Score identity Distillation: Exponentially Fast Distillation of Pretrained Diffusion Models for One-Step Generation (Nikita Starodubcev) Long and Short Guidance in Score identity Distillation for One-Step Text-to-Image Generation (Nikita Starodubcev)

03.07.24 Watch recording Access code 2. !yzK@T	Scaling the Codebook Size of VQGAN to 100,000 with a Utilization Rate of 99% (@d-shlenskii) STAR: Scale-wise Text-to-image generation via Auto-Regressive representations (Dmitry Baranchuk) ControlVAR: Exploring Controllable Visual Autoregressive Modeling (Dmitry Baranchuk)
03.07.24 Watch recording Access code VhJE6v?T	(Denis Rakitin) Extending DMD to unpaired image-to-image translation and exploring its connection to optimal transport. Regularized Distribution Matching Distillation for One-step Unpaired Image-to-Image Translation
28.06.24 Watch recording Access code LMK&J#e2	Align Your Steps: Optimizing Sampling Schedules in Diffusion Models (@michaelmar) Finite Scalar Quantization: VQ-VAE Made Simple (Denis Kuznedelev)
26.06.24 Watch recording Access code 5%CR2pJa	Interpreting the Weight Space of Customized Diffusion Models (Nikita Starodubcev) LiteVAE: Lightweight and Efficient Variational Autoencoders for Latent Diffusion Models (Denis Kuznedelev)
19.06.24 Watch recording Access code h=8c%mf0	An Image is Worth 32 Tokens for Reconstruction and Generation (@dbaranchuk) Autoregressive Image Generation without Vector Quantization (@dbaranchuk)
14.06.24 Watch recording, part 1 Watch recording, part 2	Guiding a Diffusion Model with a Bad Version of Itself (Nikita Starodubcev) Characteristic Guidance: Non-linear Correction for Diffusion Model at Large Guidance Scale (Nikita Starodubcev)
05.06.24 Watch recording Access code 518.8?qz	Visual Autoregressive Modeling: Scalable Image Generation via Next-Scale Prediction (@dbaranchuk) Muse: Text-To-Image Generation via Masked Generative Transformers (@dbaranchuk) MaskGIT: Masked Generative Image Transformer (@dbaranchuk)

31.05.24 Watch recording Access code 1MQq2^Ht	A survey of VQ-VAE/VQGAN image tokenization and autoregressive image generation. Scaling Autoregressive Models for Content-Rich Text-to-Image Generation (@dbaranchuk) Vector-quantized Image Modeling with Improved VQGAN (@dbaranchuk) Taming Transformers for High-Resolution Image Synthesis (@dbaranchuk) Generating Diverse High-Fidelity Images with VQ-VAE-2 (@dbaranchuk) Neural Discrete Representation Learning (@dbaranchuk)
29.05.24 Watch recording Access code q+iF2A=L	Recent advances in diffusion model distillation. PaGoDA : Progressive Growing of a One-Step Generator from a Low-Resolution Diffusion Teacher (Nikita Starodubcev) Improved Distribution Matching Distillation for Fast Image Synthesis (@dbaranchuk)
03.05.24 Watch recording Access code jQ&*d4pF	Fast High-Resolution Image Synthesis with Latent Adversarial Diffusion Distillation (SD3-Turbo) (@dbaranchuk) SeeSR: Towards Semantics-Aware Real-World Image Super-Resolution (Denis Kuznedelev) AddSR: Accelerating Diffusion-based Blind Super-Resolution with Adversarial Diffusion Distillation (Denis Kuznedelev)
24.04.24 Watch recording Access code hTgN2j=&	Diffusion model quantization, continued. Post-training Quantization with Progressive Calibration and Activation Relaxing for Text-to-Image Diffusion Models (@notantonvoron) TFMQ-DM: Temporal Feature Maintenance Quantization for Diffusion Models (@notantonvoron) Q-DM: An Efficient Low-bit Quantized Diffusion (@notantonvoron) Effective Quantization for Diffusion Models on CPUs (@notantonvoron)
19.04.24 Watch recording Access code eFrAil@ Passcode: &[eFrAil@] (https://sta ff.yandex-team.ru/eFrA il)	An introduction to post-training quantization of diffusion models. Efficient Quantization Strategies for Latent Diffusion Models (@notantonvoron) Ptqd: Accurate post-training quantization for diffusion models (@notantonvoron) Q-Diffusion: Quantizing Diffusion Models (@notantonvoron) Post-training Quantization on Diffusion Models (@notantonvoron) Additional source links: [1]

17.04.24 Watch recording Access code a=!6Q!Zi	CADS: Unleashing the Diversity of Diffusion Models through Condition-Annealed Sampling (@dbaranchuk) Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models (@dbaranchuk) Boosting Diffusion Models with Moving Average Sampling in Frequency Domain (@s-kastryulin)
12.04.24 Watch recording Access code AzM.61+*	Optimizing LLMs to produce better prompts for text-to-image Optimizing Prompts for Text-to-Image Generation (@notantonvoron) A User-Friendly Framework for Generating Model-Preferred Prompts in Text-to-Image Synthesis (@notantonvoron) Improving Text-to-Image Consistency via Automatic Prompt Optimization (@notantonvoron)
05.04.24 Watch recording Access code AzM.61+*	Fixed Point Diffusion Models (@michaelmar) Idempotent Generative Network (@michaelmar)
03.04.24 Watch recording Access code j=.Mr Passcode: 7A\$j[j=.Mr] (http://j=.m r/)	Structure-Guided Adversarial Training of Diffusion Models (Denis Kuznedelev) Improved Techniques for Maximum Likelihood Estimation for Diffusion ODEs (@d-shlenskii) Additional source links: [1]
20.03.24 Watch recording Access code .t?GPA1E	Foundations of 3D generation: research papers, basic 3D concepts, and neural radiance fields (NeRFs). Zero-1-to-3: Zero-shot One Image to 3D Object (@kstruminsky) DreamFusion: Text-to-3D using 2D Diffusion (@kstruminsky)
13.03.24 Watch recording Access code 0r90U\$J0	Training-free Diffusion Model Adaptation for Variable-Sized Text-to-Image Synthesis (Nikita Starodubcev) CoDi: Conditional Diffusion Distillation for Higher-Fidelity and Faster Image Generation (@dbaranchuk)
06.03.24 Watch recording Access code U*\$^yM49	How score variance at intermediate steps affects convergence speed and quality, and whether reducing it matters. Stable Target Field for Reduced Variance Score Estimation in Diffusion Models (Nikita Starodubcev) Nearest Neighbour Score Estimators for Diffusion Generative Models (Nikita Starodubcev)

28.02.24 Watch recording Access code A%Y9#&\$6	Which aspects of diffusion models are most responsible for high-level image features? Deconstructing Denoising Diffusion Models for Self-Supervised Learning (@dbaranchuk) A recap of reinforcement learning for diffusion models. Diffusion Model Alignment Using Direct Preference Optimization (@dbaranchuk)
21.02.24 Watch recording Access code dt@H4L0#	Advances in latent diffusion models for super-resolution. You Only Need One Step: Fast Super-Resolution with Stable Diffusion via Scale Distillation(@s-kastryulin) DiffBIR: Towards Blind Image Restoration with Generative Diffusion Prior (@s-kastryulin) Scaling Up to Excellence: Practicing Model Scaling for Photo-Realistic Image Restoration In the Wild (@s-kastryulin)
14.02.24 Watch recording Access code Q6#U4\$\$f	The importance of generating legible text in images with text-to-image diffusion models. Character-Aware Models Improve Visual Text Rendering (@dbaranchuk)
9.02.24 Watch recording Access code ##IFW7Pf	Analyzing and Improving OT-based Adversarial Networks (@antonplaksin) Universal Guidance for Diffusion Models (@d-shlenskii)
07.02.24 Watch recording Access code R.3V.Tj Passcode: [R.3V.Tj](http://r.3v.tj/)^	When synthetic images from diffusion models can help with representation learning and classification. Learning Vision from Models Rivals Learning Vision from Data (@dbaranchuk) StableRep: Synthetic Images from Text-to-Image Models Make Strong Visual Representation Learners (@dbaranchuk) Scaling Laws of Synthetic Images for Model Training ... for Now (@dbaranchuk) Additional source links: [1]
24.01.24 Watch recording Access code ir*VeK65	Updates on fine-tuning diffusion models with human feedback. DPOK: Reinforcement Learning for Fine-tuning Text-to-Image Diffusion Models (@shishenya) Parrot: Pareto-optimal Multi-Reward Reinforcement Learning Framework for Text-to-Image Generation (@shishenya) Directly Fine-Tuning Diffusion Models on Differentiable Rewards (@shishenya) Using Human Feedback To Finetune Diffusion Models Without Any Reward Model (@shishenya)

19.01.24 Watch recording Access code PU!Xmmd3	A detailed discussion of a generalization of consistency models. Consistency Trajectory Models: Learning Probability Flow ODE Trajectory of Diffusion (Nikita Starodubcev)
17.01.24 Watch recording Access code J##=5c6!	A closer look at consistency models (the second paper was not covered in detail due to time constraints). Consistency Models (Denis Kuznedelev) Improved Techniques for Training Consistency Models (Denis Kuznedelev)
12.01.24 Watch recording Access code tm7s+*SL	Continuation of the previous seminar. Concluding the discussion of diffusion inversion with guidance (Nikita Starodubcev) Generalizing diffusion processes to different prior distributions. Star-Shaped Denoising Diffusion Probabilistic Models (@antonplaksin)
10.01.24 Watch recording Access code wv&#A*3y	Image editing through inversion of text-to-image diffusion models. LEDITS++: Limitless Image Editing using Text-to-Image Models (Nikita Starodubcev) Negative-prompt Inversion: Fast Image Inversion for Editing with Text-guided Diffusion Models (Nikita Starodubcev) An Edit Friendly DDPM Noise Space: Inversion and Manipulations (Nikita Starodubcev)

2023

29.12.23 Watch recording Access code 5?r5ZV@f	In-context learning in text-to-image diffusion models Context Diffusion: In-Context Aware Image Generation (@dbaranchuk) In-Context Learning Unlocked for Diffusion Models (@dbaranchuk)
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